

MedVAE: adaptation, fine-tuning and analysis for medical imaging and coronary segmentation

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Context. Deep learning in medical imaging is limited by the high resolution required for diagnostic detail. VAEs compress the image into a compact latent.

- **MedVAE:** generic medical compressor ($> 1\text{M}$ images), $\times 4$ per side, $\times 16$ fewer values ($512^2 \rightarrow 128^2 \times 1$), trained in two stages (reconstruction then semantic enrichment).

Domain. Coronary angiography (ARCADE, 1 500 images): thin, tubular, low-contrast structures, noise and artefacts.

Central question

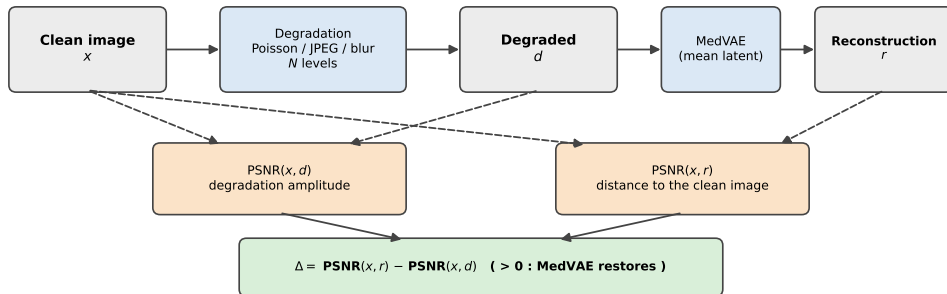
Does a generic medical compressor remain **reliable and useful** when facing a specific and demanding clinical domain?

Four complementary axes:

- 1 Robustness to degradations
- 2 Image-quality inductive bias via FiLM
- 3 Semantic structuring of the latent (JEPA)
- 4 Latent representation for segmentation

Robustness of MedVAE to input degradations

Method: evaluation protocol



The reconstruction is compared to the **clean** image: does MedVAE bring a degraded input closer to it? 100 seg_val images, deterministic latent, fixed intensity scale, seeded noise, bootstrap CIs.

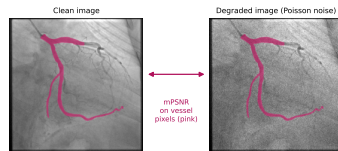
Why not a no-reference metric?

Discarded approach: score the input quality with ARNIQA (no-reference), then correlate it with the reconstruction quality.

ARNIQA — clean dataset (n=1000)

min / max	0.2695 / 0.6512
mean $\pm$ std	0.4570 $\pm$ 0.0482
quantiles 5/50/95	0.3772 / 0.4566 / 0.5391

Weak correlation with the reconstruction PSNR (ARNIQA: $r = -0.37$, wrong sign; custom score: $r \approx -0.01$): the quality variance between clean images is too small to serve as an exploitable axis.



mPSNR illustration (vessel mask)

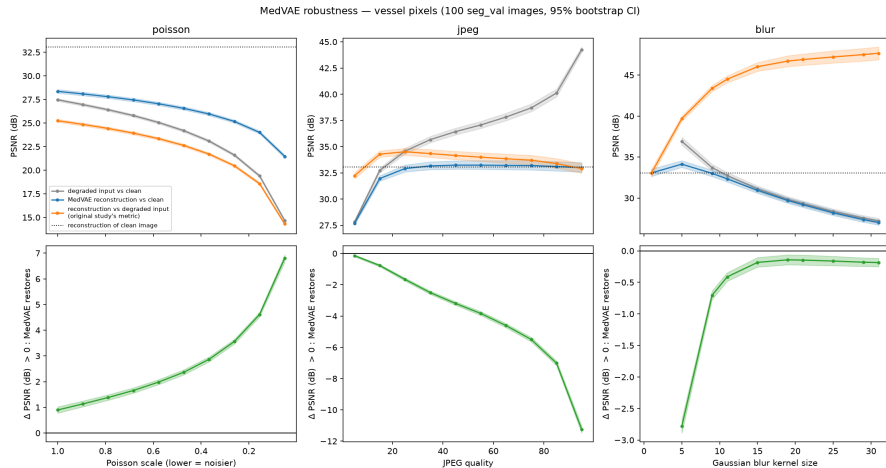
Result: MedVAE reduces noise (less than a simple filter), not blur or JPEG

Degradation	Δ PSNR (dB)	Restored
Poisson	+1.2 $\rightarrow$ +6.6	99–100%
5×5 filter	+6.3 $\rightarrow$ +10.3	100%
JPEG (95 $\rightarrow$ 5)	−11.7 $\rightarrow$ −0.5	0–9%
Blur	−2.0 $\rightarrow$ +0.1	–

$$\Delta = \text{PSNR}(\text{clean}, \text{recon}) - \text{PSNR}(\text{clean}, \text{degraded})$$

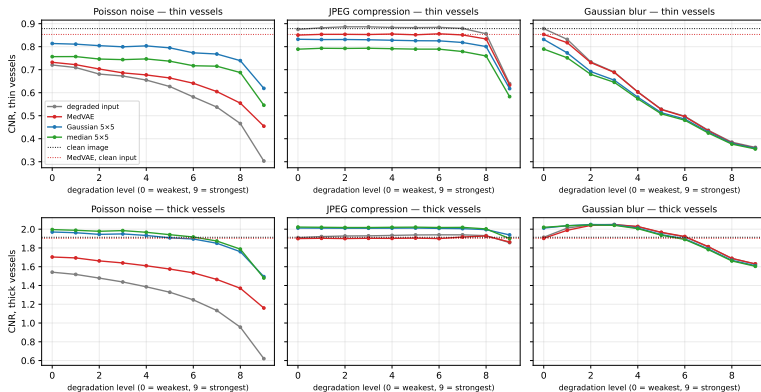
- **Poisson**: noise reduced, but 3.7–5.6 dB less than a simple 5×5 Gaussian filter
- **JPEG**: never restored — MedVAE’s own error (33 dB ceiling) exceeds the artefacts
- **Blur**: not restored — the reconstruction follows the blurred input
- The earlier “blur reconstructs better” measured fidelity to the *degraded* input (orange)

Result: restoration curves (vessel pixels)



Grey: degraded input vs clean. Blue: reconstruction vs clean. Orange: reconstruction vs degraded input (earlier metric). Bottom: Δ .

Vessel visibility: contrast-to-noise ratio (CNR)

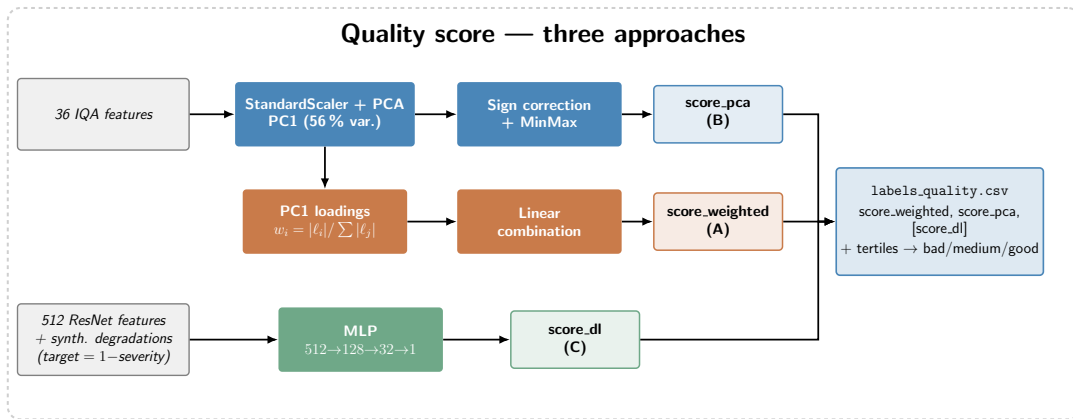


- $CNR = (\mu_{bg} - \mu_{vessel}) / \sigma_{bg}$, thin (≤ 6 px) and thick (≥ 12 px) vessels
- **Poisson:** MedVAE raises the CNR (thick $+0.16 \rightarrow +0.54$, thin $+0.01 \rightarrow +0.15$), but the 5×5 filter does better, **thin vessels included**
- Without noise (JPEG, blur): MedVAE leaves the CNR unchanged; next step: detectability d' of a model observer

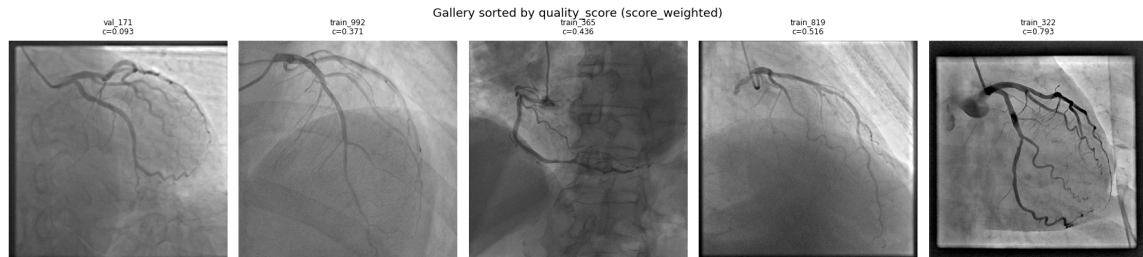
Image-quality inductive bias (FiLM)

- MedVAE applies the **same transform** to a clean or a heavily degraded coronary angiography: no notion of *how degraded* the input is.
- **Hypothesis:** conditioning MedVAE on a quality score $c \in [0, 1]$, injected via FiLM, should let it adapt to the degradation level actually present.
- Three independent ways to compute c (A: weighted, B: PCA, C: learned): in-depth study with C (3 seeds), checks with A and B.

Building the quality score c

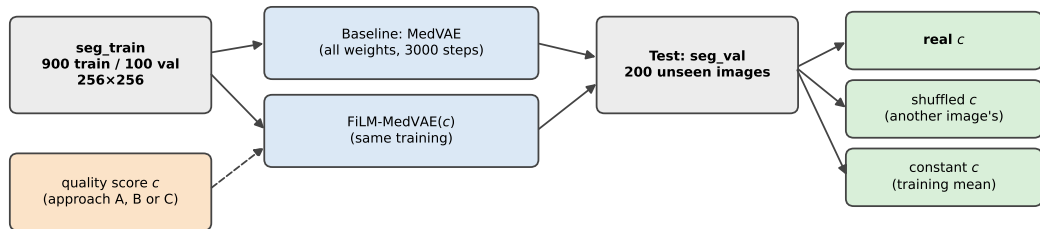


Does the score make clinical sense?



ARCADE images sorted by increasing score c (approach A).

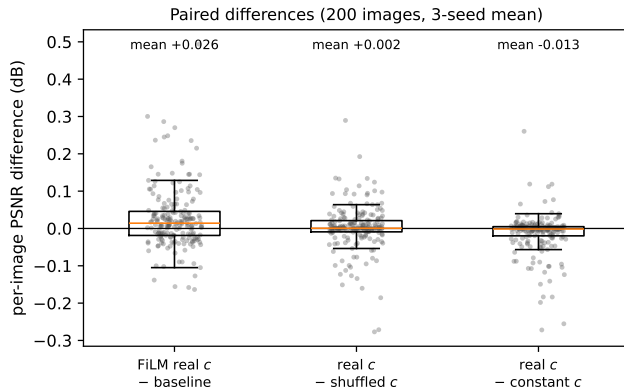
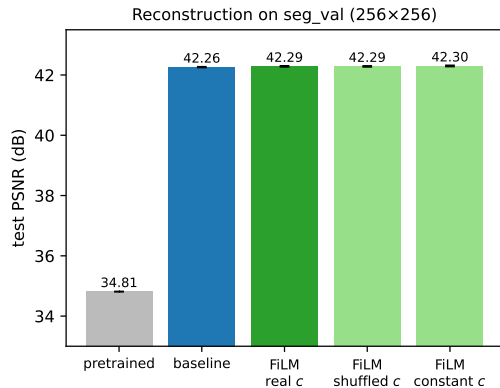
FiLM conditioning & revised protocol



*PSNR, vessel PSNR, SSIM, HaarPSI · 3 seeds · paired per-image differences
If the model uses c: real c > shuffled c and real c > constant c*

- FiLM after each of the 19 ResNet blocks: $h \leftarrow (1 + \gamma(c)) h + \beta(c)$, zero-initialised (+1.7M parameters, +3 %)
- **Revised:** 256×256 (was 64×64 , where MedVAE only reaches 20.7 dB), KL normalised per pixel, same training for both models, 3 seeds

Result: fine-tuning helps, conditioning does not



- Fine-tuning MedVAE on ARCADE: **+7.4 dB** (vessels: +5.3 dB), same for all seeds
- FiLM vs baseline: +0.03 dB — negligible

Does the network use c ?

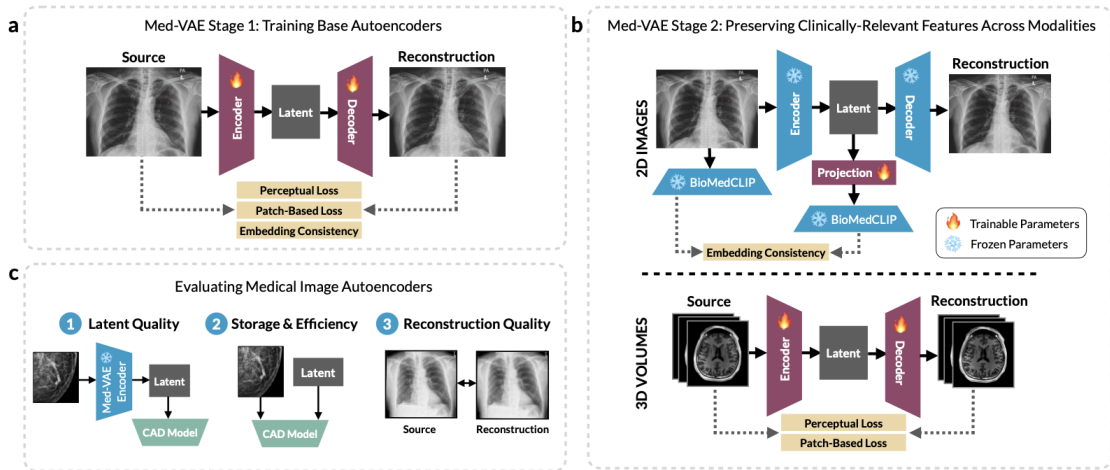
PSNR (dB), 3 seeds	true — shuffled c	FiLM — base
Score C	+0.002	+0.03
Score A	+0.007*	+0.04
Score B	+0.033*	+0.05

* CI excluding 0, same sign for every seed; FiLM — baseline: CI over seeds includes 0

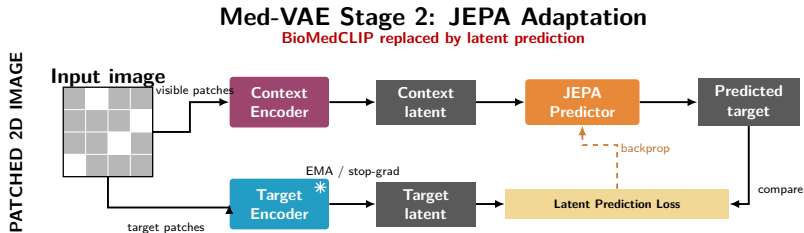
- **Score C: no** — it barely varies in training (79 % above 0.9)
- **Scores A, B: yes, but barely** (≤ 0.03 dB)
- FiLM gains come from the added parameters, not from the quality information
- c is computed *from the image*: for an autoencoder of that image, it carries nothing new
- The earlier “only C is exploited” compared two separately trained single runs

Semantic structuring of the latent space (JEPA)

Original stage 2 — BioMedCLIP distillation



Modified stage 2 — JEPA adaptation



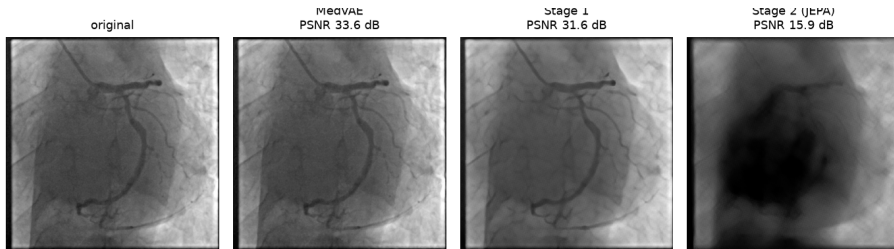
$$\mathcal{L}_{\text{stage2}} = \mathcal{L}_{\text{pred}} + \lambda_{\text{SIGReg}} \mathcal{L}_{\text{SIGReg}}$$

$$\mathcal{L}_{\text{pred}} = \frac{1}{|\mathcal{M}|} \sum_{p \in \mathcal{M}} \text{SmoothL}_1 \left(\hat{z}_t^{(p)}, \tilde{z}_t^{(p)} \right), \quad \mathcal{L}_{\text{SIGReg}} = \mathbb{E}_{\mathbf{v}} \left[T_{\text{EP}} \left(\{z_c^{\top} \mathbf{v}\} \right) \right]$$

- **Stage 1 is not MedVAE**: same family, but smaller (21M), $\times 8$ per side, trained *from scratch* on ARCADE
- **Earlier issues**: test images in the pre-training; the “25 dimensions for 90 % of the variance” measured `channel_proj`, a layer *never trained*
- Now: retrained without the test set, **3 pre-trainings**; evaluated on the layer JEPA trains (features before `conv_out`), 300 test images; probe lr chosen on validation

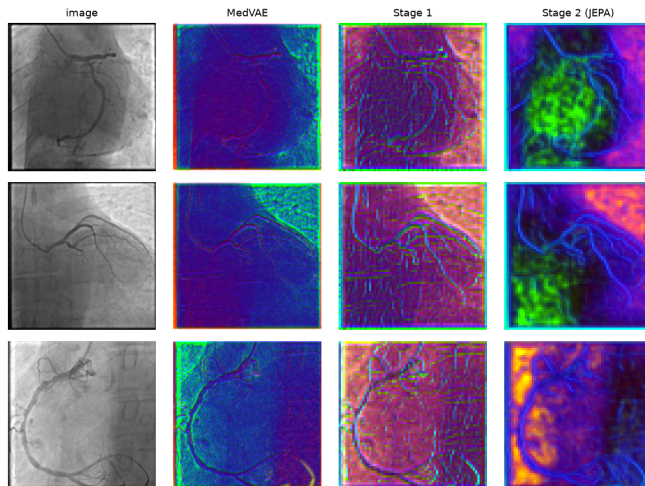
	MedVAE	Stage 1	Stage 2 JEPA
PSNR (dB)	34.8	31.1	18.6
Vessel linear probe (Dice)	0.517	0.561	0.536
Effective rank / dims for 90 % var.	251 / 61	147 / 46	156 / 50

Results — reconstruction and features



- **Frozen** decoder: the adapted encoder is no longer decodable (18.6 dB)
- JEPA **lowers** vessel separability vs stage 1 (-0.025 , 3 pre-trainings out of 3), ends level with MedVAE; **no contraction** (rank 147 $\rightarrow$ 156, no consistent direction)
- Best features: stage 1, before JEPA

Results — RGB PCA of the intermediate features



Stage 2 features: large homogeneous regions, vessels barely visible.

Integration for multi-class segmentation

Task. Fine segmentation of coronary arteries on ARCADE:

- 26 classes (the background and 25 segments)
- thin, tubular, low-contrast structures
- extreme imbalance ($\approx 95\%$ background)

Tool. MedVAE, a pre-trained medical variational autoencoder: $\times 16$ compression in number of values ($512^2 \rightarrow 128^2 \times 1$), but **lossy**.

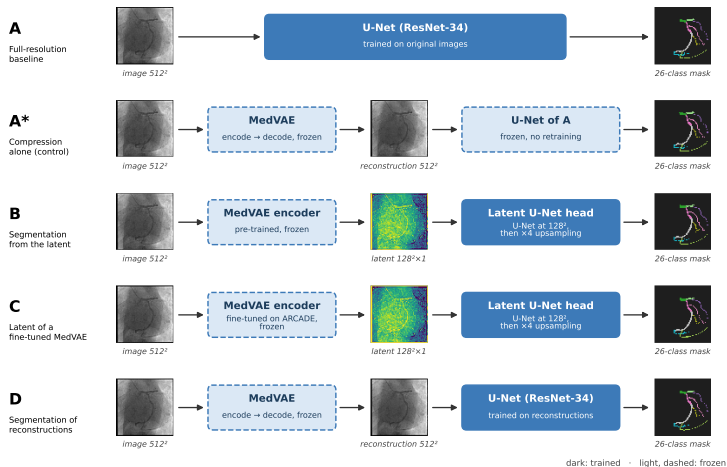
Two questions

- 1 Does the compression preserve enough anatomical information for segmentation?
- 2 Where to integrate it: **latent** space or **pixel** space?

Method: five conditions and two controls

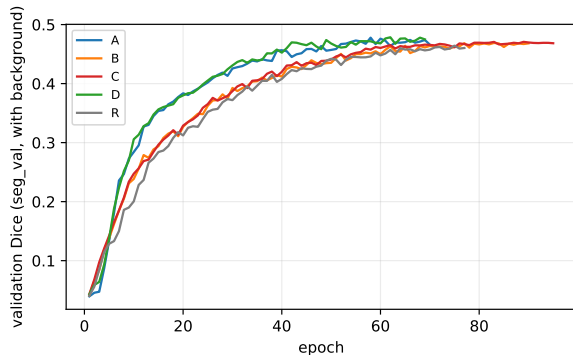
- **A** — full-resolution U-Net
- **A*** — U-Net of A on reconstructions, **no retraining**
- **B** — U-Net head at the latent resolution, frozen MedVAE encoder
- **C** — like B, MedVAE fine-tuned on ARCADE
- **D** — U-Net retrained on reconstructed images
- **R, A^{↓↑}** — controls: 4×4 average pooling instead of MedVAE

3 seeds per condition; artery Dice (background excluded) on the 300 test images.



Result 1: the latent works, like a downsampled image

- B: **0.411**, C: **0.410** artery Dice, i.e. **96 %** of A (0.428), from a $16\times$ smaller input
- **But** the same head on the image downsampled to 128×128 (R): **0.405** (B - R = +0.006, not established)
- Earlier “the latent fails” (Dice 0.04) was a training issue:
 - head learning rate 5×10^{-5} , too low $\rightarrow 10^{-3}$
 - no convolution at the latent resolution $\rightarrow$ small U-Net head
- Fine-tuning MedVAE (C): no effect (C - B = -0.001)

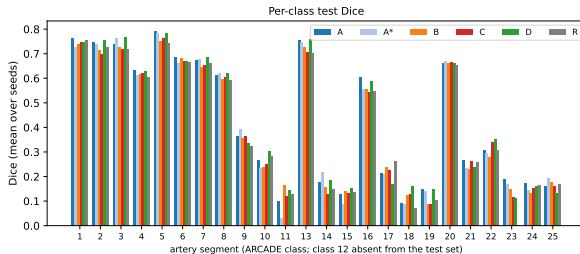
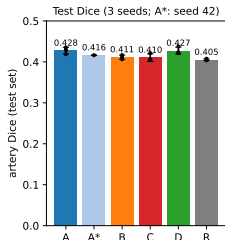


Validation Dice, mean over 3 seeds

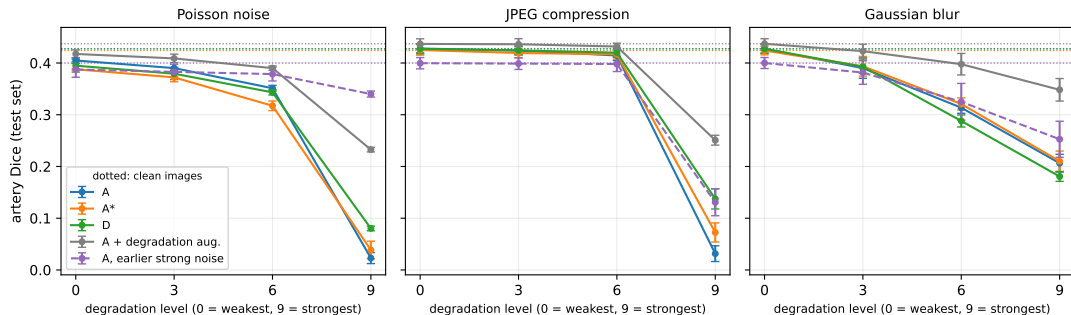
Result 2: pixel space preserves, adaptation adds nothing

Cond.	Artery Dice	Artery IoU
A	0.428 ± 0.008	0.309
A*	0.424 ± 0.008	0.305
B	0.411 ± 0.005	0.292
C	0.410 ± 0.009	0.291
D	0.427 ± 0.009	0.309
R	0.405 ± 0.002	0.288
A $\downarrow\uparrow$	0.389 ± 0.018	0.275

- **Compression alone costs little:** $A^* \approx A$ (-0.004), whereas naive downsampling costs -0.039 ($A^{\downarrow\uparrow}$)
- **D $\approx$ A** (0.000, within seed variability)
- **Earlier “D gains on thin segments” (+0.017):** gone with 3 seeds and the corrected augmentation

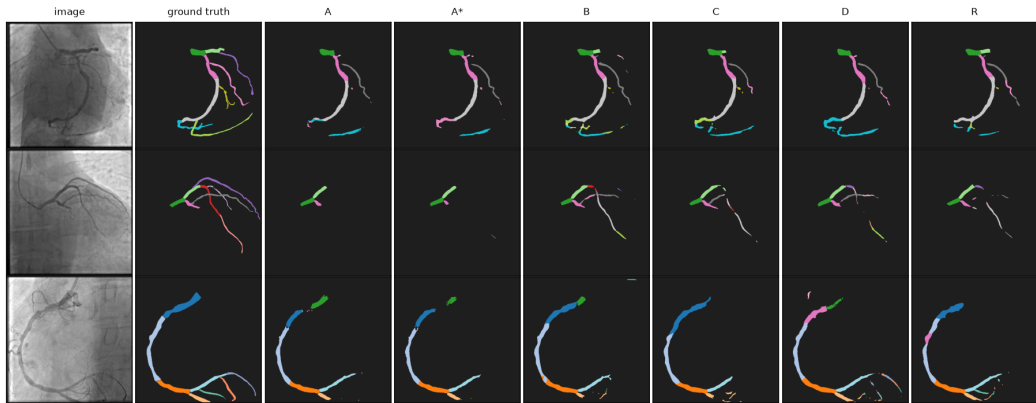


Segmentation of degraded images



- MedVAE's noise reduction does **not** help: $A^* \leq A$ under Poisson noise, $D \approx A$
- Robustness comes from training: realistic degradations in A's augmentation $\rightarrow$ strongest Poisson $0.02 \rightarrow 0.23$, JPEG $0.03 \rightarrow 0.25$, blur $0.21 \rightarrow 0.35$, at no cost on clean images (same degradation families as in training)

Qualitative predictions



Seed-42 models on three test images: A, A* and D nearly identical; B, C (latent) and R (downsampled image) follow the coronary tree at 128×128 .

General conclusion

Across four complementary axes, we probed the behaviour of MedVAE, a generic medical compressor, on the demanding domain of coronary angiography:

- **Robustness:** reduces Poisson noise (less than a simple filter), restores neither JPEG nor blur — and does *not* make segmentation more robust.
- **FiLM:** fine-tuning on ARCADE gains 7 dB; conditioning on a quality score adds nothing, the score being a function of the image itself.
- **JEPA:** with a frozen decoder, degrades reconstruction and vessel separability; no dimensional contraction.
- **Segmentation:** $\times 16$ compression keeps the anatomy in pixel space ($A^* \approx D \approx A$, unlike naive downsampling); the latent reaches 96 % of A's Dice, but so does a downsampled image.

Takeaway. A faithful compression, reusable in pixel space; its latent is usable, not better than downsampling; several first-version conclusions reversed once the protocol was fixed (right reference, same training, several seeds, trained layer).

Toward a *foundation model* for radiation-based medical imaging

Unify the explored axes into a single training objective.

- **Hybrid JEPA + reconstruction objective:** predict the masked latent patches while keeping the encoder anchored to a decodable space.
- **Physically grounded degradations:** quantum noise, contrast/brightness, multiplicative fields, anisotropic motion blur (cardiac/respiratory).
- **Conditioning (CVAE)** on DICOM metadata *not* derived from the image (modality, dose, acquisition parameters) → disentangled content / acquisition-quality representations.
- **Downstream validation:** classification, detection, multi-modality segmentation (ARCADE as a demanding test bed).